

NET724 EVO

Programmable control board Operating instructions and warnings

Index

1	Product Description	31	7	Complete parameter list	39
2	Technical Data	31	8	Detailed parameter list	42
3	Electrical Connections	32	9	EVO Menu Display	54
4	Programming	34	10	"EASY" parameter list	55
5	Description of Inputs / Outputs	35	11	Commissioning	56
6	Messages shown on the Display	38	12	Product Disposal	56

SYMBOLS

The following symbols are used within this manual to indicate potential hazards.

	Important safety notice. Failure to follow these instructions may result in serious injury or damage to property. Failure to do so may result in product malfunction and create a hazardous situation.
	Important safety notice. Contact with live parts can result in death or serious injury.
	Important information for installing, programming or commissioning the product.

1 PRODUCT DESCRIPTION

NET724 EVO is a control unit for **DEA** System automations with a single 24V motor.

The main feature of this control unit is the simplicity of the input and output configuration, which makes it adaptable to any type of automation as required. All that is required is to set the desired configuration for the automation being used to have the operating parameters optimally set, excluding all other unnecessary functions.

2 THECNICAL DATA

Supply voltage (V)	230 - 240 V ~ (50/60 Hz)	
Transformer rated power (VA)	80 VA (230/25V)	150 VA (230/25V)
Fuse F1 (A)	Fuse T 1A 250V	Fuse T 2A 250V
Auxiliary devices power supply output	24 V --- max 200mA	
Output 1 configurable	24 V --- max 5 W	
Output 2 configurable	24 V --- max 5 W	
Radio receiver frequency	433,92 MHz	
Type of remote control encoding	HCS fix-code - HCS rolling code - Dip-switch - DART	
No. of remote controls managed	200	

3 ELECTRICAL CONNECTIONS



! Risk of injury and material damages due to electric shock !



! Risk of malfunction due to improper installation !

Make the connections by following the indications of the wiring diagram.

WARNING For adequate electrical safety, all cables must have double insulation. Be sure to keep them clearly separated (**minimum 4 mm in air or 1 mm through additional insulation**) safety extra-low-voltage cables from low-voltage cables (230V ~) by arranging them within plastic raceways and securing them with appropriate clamps near the terminal blocks.

WARNING For the connection to the network, use multipolar cable of minimum section 3x1,5mm² and the type required by current regulations. To connect the motors, use cable minimum section 1.5 mm² and the type required by current regulations. By way of example, if the cable is outside (outdoors), it must be at least equal to H05RN-F while, if inside (in channel), it must be at least equal to H05VV-F.

WARNING Connect to the mains 230 - 240 V ~ 50/60 Hz via an omnipolar switch or other device that ensures omnipolar mains disconnection, with a contact opening distance = 3 mm.

WARNING All cables should be stripped and unsheathed in the immediate vicinity of the terminals. Keep the cables slightly longer so that any excess can be removed later.

WARNING Connect the grounding conductor to the appropriate terminal, taking care to keep its length longer than that of the active conductors so that in the event of the cable coming out of the fixing seat, the active conductors are the first to stretch.

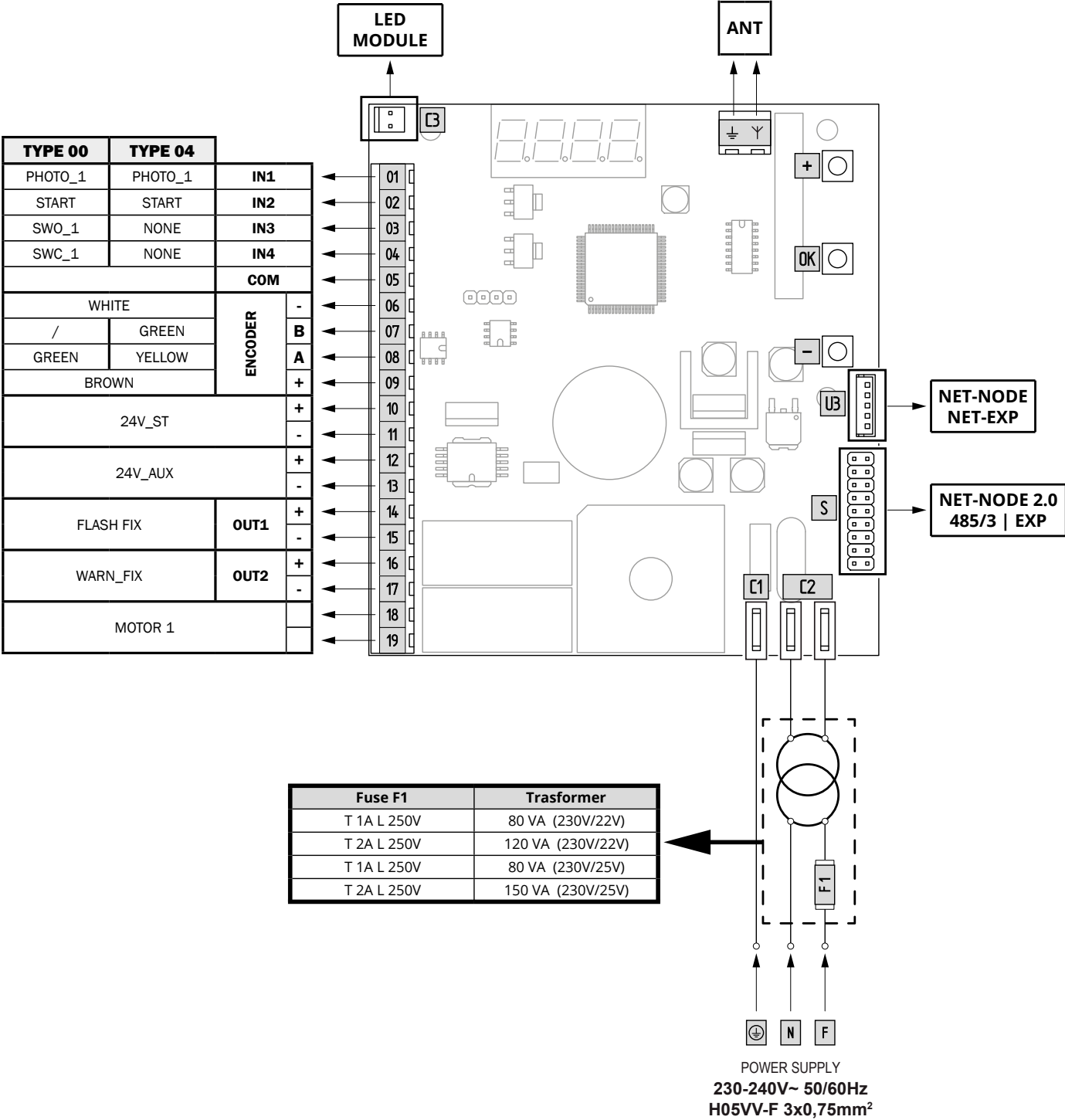
WARNING When connecting the encoder to the control board, use only a dedicated 3x0,22mm² cable.

WARNING If poor radio range is experienced, it is recommended to connect the antenna of the flashing light (if present) or to install a tuned external antenna.

NET724 EVO board terminal table

1	Input IN1		
2	Input IN2		
3	Input IN3		
4	Input IN4		
5	Inputs common		
6	-	M1 motor encoder output	
7	B		
8	A		
9	+		
10	+	Stabilized 24V === power supply output for tested safety devices	(24V_ST + 24V_AUX) = max 200mA
11	-		
12	+	24V === power supply output for auxiliary devices (e.g. BAT_ADV accessory)	
13	-		
14	+	OUTPUT 1 configurable 24V === max 5W (see P066 Io.31 for the selectable values)	
15	-		
16	+	OUTPUT 2 configurable 24V === max 5W (see P052 Io.32 for the selectable values)	
17	-		
18-19	M1 max 5A motor output		
ANT	↯	Radio antenna signal input	
	⏏	Radio antenna earth input	
C 1	Motor earth terminal		
C 2	Input for 25V ~ power supply from transformer		
C 3	Output for courtesy LED light module connector		
S	NET-NODE 2.0 - 485/3-EXP module connector input		
U 3	NET-NODE - NET-EXP plug-in module connector		

WIRING DIAGRAM
NET 724EVO



Description	Cable	Length	
		(1m / 20m)	(20m / 50m)
24 V power supply	HAR EN50575 CPR 305/2011	3 x 1,5 mm²	3 x 2,5 mm²
230 V power supply		4 x 0,5 mm²	4 x 2,5 mm²
Warning light		2 x 0,5 mm²	2 x 1,0 mm²
TX photocell		2 x 0,5 mm²	2 x 1,0 mm²
RX photocell		4 x 0,5 mm²	4 x 1,0 mm²
Key-operated selector		3 x 0,5 mm²	3 x 1,0 mm²
Encoder	RG58	3 x 0,5 mm² (max 15m)	
Antenna		max 20m	

4 PROGRAMMING

4.1 Power supply

Power the device. On the display the figures/words **"00.01"** (or the firmware version currently being used), **"TYPE"**, **"-00-** (or the value of the TYPE of operation currently being used) will appear in sequence followed by the gate closed symbol **"- - -"** (see Table "STATUS MESSAGES on page 38").

NOTE: The sequence of messages appearing on the display can be integrated, after the control unit is powered, with the information relative to the number of total manoeuvres completed up to that time. Consult parameter P151|EX.18 to activate or deactivate the function.

4.2 Setting or changing the TYPE

If the TYPE set is not the desired one or if it has not been defined (“-88-” flashing), proceed as follows:

1. With the door closed, press and hold the [OK] button, then simultaneously press and hold the [+] and [-] buttons until the message **"TYPE"** appears;
2. Release all 3 keys; The flashing number corresponds to the currently set TYPE.

If the set value is correct, exit the procedure by pressing the **[+]** and **[-]** keys simultaneously.

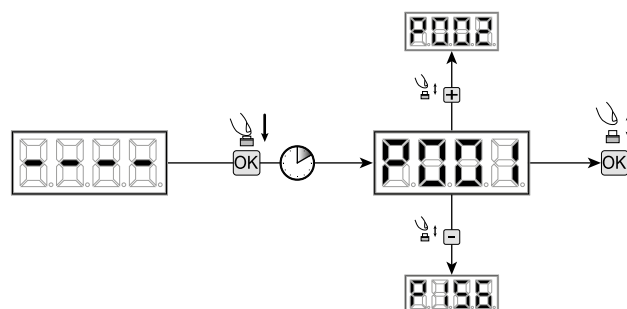
3. If you want to change the TYPE, scroll through the selectable values with the [+] or [-] keys and at the end confirm the selection by keeping the [OK] key pressed.

NOTE: To confirm that the TYPE has been changed, the display will light up all the segments and will restart the control unit, showing in sequence the words “00.01” (or the firmware version currently in use), “TYPE”, “-00-” (or the operating TYPE value currently in use) followed by the closed gate symbol “- - - -”.

4.3 Accessing the programming menus

NOTE: Make sure parameter P156|TE.11 is set to 000.

With the door stopped, press and hold down the [OK] key until **"P001"** appears:



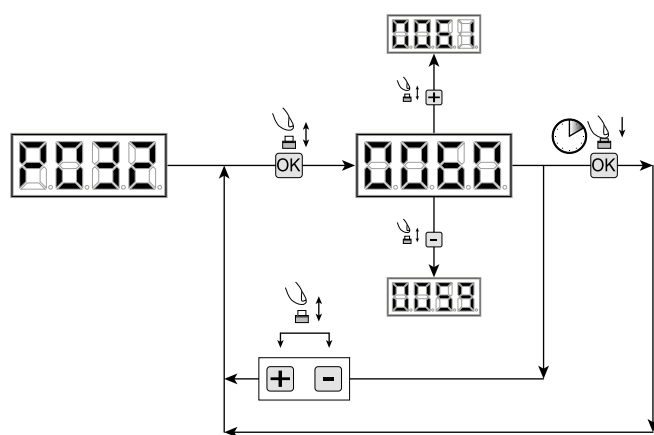
4.4 Navigation and modification of parameters

1. With “**P001**” displayed on the screen, use the [+] and [-] buttons to scroll through the parameters until the desired one is reached (e.g., P032).
2. Access the parameter using the [OK] button.
3. Use the [+] and [-] buttons to set the desired value and confirm the selection by holding down the [OK] button.

NOTE: To exit a parameter or go back to a previous step, simultaneously press the [+] and [-] buttons.

NOTE: Some parameters may not have selectable values but will initiate procedures that need to be followed as indicated in the detailed parameter list (page 42).

It is possible that some parameters are visible but not selectable. Always ensure that the parameter is actually usable depending on the set TYPE.



5 DESCRIPTION OF INPUTS / OUTPUTS

The tables shown below describe the operation of all possible selections for both the inputs and outputs present on the board.

INPUTS (IN / EXP_IN)	
Message	Description
NONE	Not used
START	N.O. input. Start. Triggers an opening or closing manoeuvre. It can function in “reversal” mode (P049 EX.06=0), “step-by-step” mode only with START (P049 EX.06=1) or “step-by-step” mode with START, OPEN and CLOSE (P049 EX.06=2).
PED	N.O. input. Pedestrian. Triggers the partial opening of the gate. The pedestrian opening distance can be adjusted through P043 Mo.12.
OPEN	N.O. input. Open. Triggers the opening of the gate.
CLOSE	N.O. input. Close. Triggers the closing of the gate.
OPEN_PM	N.O. input. Hold-to-run open. The gate opens for the time that the button is kept pressed.
CLOSE_PM	N.O. input. Hold-to-run close. The gate closes for the time that the button is kept pressed.
OPEN_INT	Internal OPEN command, causes the internal traffic light to switch when the gate reaches its open position. If an OPEN_EXT command is also received before the gate starts to close it will process this before the automatic close time begins.
OPEN_EXT	External OPEN command, causes the external traffic light to switch when the gate reaches its open position. If an OPEN_INT command is also received before the gate starts to close it will process this before the automatic close time begins.
OPEN_STOP	N.O. input. Opens the gate and stops it (the TCA is disabled). At the end of the opening manoeuvre, the board enters the STOP mode until the next START or CLOSE command is given.
AUX_1_IN	Input for commanding the AUX_1_OUT output.
AUX_2_IN	Input for commanding the AUX_2_OUT output.
COURTESY_IN	N.O. input for commanding the COURTESY output. Triggers the output for the duration of the COURTESY Time (P130 tl.17). If the output is already active, a new COURTESY_IN command reloads the courtesy time.
STOP / SAS_INPUT	N.C. contact. (SAS_INPUT): If connected to WARN_FIX / SAS_OUTPUT on a second control unit, it triggers the “bank door” mode (disabling of the opening of the second door until the first has re-closed completely). N.C. stop input. It stops the movement during any manoeuvre. If not used, bridge the input.
SWO_1	N.C. input. 1 motor opening limit switch. If not used, disable the input with the relative parameter.
SWC_1	N.C. input. 1 motor closing limit switch. If not used, disable the input with the relative parameter.
PHOTO_1	N.C. input. Photocell 1. To select the operating mode, see P050 SA.01. If not used, bridge the input.
PHOTO_2	N.C. input. Photocell 2. To select the operating mode, see P051 SA.02. If not used, bridge the input.
SAFETY_1	N.C. input. Safety edge 1. To select the operating mode, see P067 SA.03. If not used, bridge the input.
SAFETY_1_8k2	Analogue input for safety edge 1 with 8k2.
SAFETY_2	N.C. input. Safety edge 2. To select the operating mode, see P068 SA.04. If not used, bridge the input.
SAFETY_2_8k2	Analogue input for safety edge 2 with 8k2.
SAFETY_INHIBITION	N.C. input. SAFETY inhibition. When open, it triggers the bypassing of the SAFETY inputs which are ignored even if active.
EMERGENCY_IN	N.C. contact. If open, it causes total opening and maintenance in the open position until the contact is closed again. The emergency opening can be temporarily interrupted by: - closing, START, STOP or photocell commands: movement stop; - obstacle detection intervention: reversal of motion (partial or complete). After these interruptions, the emergency opening resumes anyway as long as the EMERGENCY_IN command remains active. After the reclosing of an EMERGENCY_IN type contact, any automatic reclosing (if enabled) is performed or ignored according to the TCA parameter with EMERGENCY_IN (P127 SA.09). Note: If the control panel is in “RESP” status, emergency opening can only occur if the parameter P053 Mo.17 is set to values 000, 002, or 004 (reference search on opening). Otherwise, it will be necessary to allow the “RESP” status to conclude in closing before activating the EMERGENCY_IN function. DO NOT USE EMERGENCY_IN IF THE INSTALLATION PROVIDES THE USE OF ONLY DEAD-MAN COMMANDS FOR OPENING.
RESET	N.C. contact. Triggers the resetting of the control unit.
MANEUVER_INHIBITION	N.C. contact for inhibition of opening and closing operations For the selection of the operating mode see P153 EX.20. If not used, bridge the input.

OUTPUTS (OUT / EXP_OUT)	
Message.	Description
NONE	Not used
24V	Output for 24V === power supply for accessories. This setting can only be used on outputs that are powered.
24V_TEST	Output for 24V === power supply for controlled safety devices (the output is switched off during the test if the safety device auto-test is enabled (P071 SA.10). Use this type of output also for checking the switching off of accessories with the energy-saving function active (P061 EX.10). This setting can only be used on outputs that are powered. In the case of control of safety devices, these must be wired and aligned before learning the stroke (P003 SE.03).
ELOCK_M1	N.O. contact. Output for M1 motor electric lock.
ELOCK_INV_M1	N.O. contact. Output for M1 motor inverted electric lock (for example for operating the electromagnet of the barriers).
ELOCK_BOOST_M1	“Boost” power supply output for powering the DEA electric lock item 110. This setting can only be used on outputs that are powered.
WARN_FIX SAS_OUTPUT	N.C. contact. (SAS_OUTPUT): If connected to STOP / SAS_INPUT on a second control unit, it triggers the “bank door” mode (disabling of the opening of the second door until the first has re-closed completely). N.O. contact. (WARN_FIX): Operates as a steady open gate indicator.
WARN_INT	Warning light gate indicator. Slow flashing output during opening and fast during closing, always ON with gate open, always OFF only at the end of a closing manoeuvre.
FLASH_FIX	N.O. contact. Steady warning light output.
FLASH_INT	N.O. contact. Intermittent warning light output.
COURTESY	Output for controlling the courtesy light. The output is active while the motors are running plus an additional time governed by the COURTESY time parameter (P130 tl.07). This setting can only be used on outputs that are powered.
EBRAKE_M1_N.ON	Output for electric parking brake control on reversible motors (e.g. REV/RF). The brake is energized (activated) with the motor stopped and in the presence of power. During the maneuver or in the absence of power, the output remains open and the brake is de-energized.
EBRAKE_M1_N.OFF	Output for electro-stop brake control (anti-inertia function, e.g. GULLIVER/N/3P/F/M). The brake is only energized (released) during the manoeuvre. In all other circumstances the brake is de-energized and brakes the motor.
MINUTERIE	N.O. contact. The contact closes for 3 seconds at the start of each manoeuvre.
ALARM	N.C. contact. The contact always remains open and closes when a manoeuvre fails to start due to a safety input (PHOTO, SAFETY, STOP) being active. The contact opens again when a subsequent attempt to start a manoeuvre succeeds. If there is no power, the contact is closed and can thus be used to generate an alarm.
AUX_1_OUT_INPULS AUX_2_OUT_INPULS	N.O. contact. Output controlled by the AUX_1_IN / AUX_2_IN input in impulse mode.
AUX_1_OUT_STEP AUX_2_OUT_STEP	N.O. contact. Output controlled by the AUX_1_IN / AUX_2_IN input in step-by-step mode.
AUX_1_OUT_TEMP AUX_2_OUT_TEMP	N.O. contact. Output controlled by the AUX_1_IN / AUX_2_IN input in timer-controlled mode (the value set through parameters P138 tl.16 and P139 tl.17 indicates the switch-off delay in seconds).

TRAFFIC_LIGHT_INT TRAFFIC_LIGHT_EXT	To ensure correct switching between the red and green lights on traffic light units, the output must have a switching relay (e.g. NET-EXPANSION outputs). The N.C. contact of the relay controls the red light, while the N.O. contact controls the green light. If the output is not of this type but is digital, it can be used to control traffic lights through an additional switching relay (not supplied).	
	GENERAL BEHAVIOUR	
	With the automation closed and/or the motors moving, or during the pre-manoeuve flashing, the traffic light is red. The traffic light turns green only once the opening manoeuvre has been completed. During a position reset procedure (stop/limit switch search), any traffic light remains red until the manoeuvre has ended.	
	TRAFFIC LIGHTS WITHOUT PRIORITY	
	Use the OPEN command to open the automation, set the automatic closing TCA (P041 tl.01). All traffic lights will work simultaneously in the same way by following the general behaviour described above.	
STATUS_1 STATUS_2	TRAFFIC LIGHTS WITH PRIORITY	
	Use the OPEN_INT and OPEN_EXT commands on the opposite sides of the gate. Determine the time required for completing the full path between the 2 traffic lights (clearance time). Set an automatic closing time TCA equal to double the required clearance time.	
	The traffic lights will behave as follows:	
	With the automation closed both traffic lights will be red. When a command is received from a direction (INT or EXT), it becomes a "priority". At the end of the opening manoeuvre, only the corresponding "priority" traffic light (EXT or INT) turns green, the other one stays red. In the absence of other commands, the "priority" traffic light stays green for half the TCA and then turns red. At the end of the TCA, the automation closes. If, while the "priority" traffic light is green, new commands arrive from the same direction, the TCA is reloaded and then the green light. Any command in the opposite direction is recorded and put in stand-by. This prevents the formation of any vehicle codes.	
	Once the "priority" traffic light turns red, both traffic lights remain red for the time required (TCA/2) to clear the access point. At this point, the command previously recorded on the opposite side becomes the "priority" causing its traffic light to turn green and the TCA to restart.	
STATUS_1 STATUS_2	Output for monitoring a few of the automation's statuses (see parameters P093 lo.60 and P094 lo.61). N.O. contact: closes when the monitored status occurs.	
	* Please note: for certain automation statuses to be monitored, it is necessary to have wired and correctly functioning limit switches.	
	AVAILABLE STATUSES	
	MAINTENANCE	Activates the maintenance signal
	PHOTO	Activates during the opening of a PHOTO-type input
	STOP	Activates during the opening of a STOP-type input
	OBSTACLE	Activates when an obstacle is detected by the anti-crushing function or a SAFETY-type input
	EMERGENCY	Active during activation of the EMERGENCY_IN type input
	* CLOSED_M1	Activates if M1 is in the closed status
	* OPENED_M1	Activates if M1 is in the open status
	RESP_FIX	Activates after blackout or control board reset and RESP not completed. Fixed output
	RESP_INT	Activates after blackout or control board reset and RESP not completed. Intermittent output
	* CLOSURE_FAILED	Activates if the door is not closed after the duration set in the P141 tl.19 parameter. If the automation subsequently completes the closing manoeuvre, the output deactivates. Note: Functions regardless of the presence of the TCA (P041 tl.01) or TCA_PED (P042 tl.02) active.
	SECURITY	Activates when, with the motor off, the encoder detects an unexpected movement

6 MESSAGES SHOWN ON THE DISPLAY

STATUS MESSAGES		
Mess.	Description	
----	Gate closed	
— — —	Gate open	
— — —	Pedestrian opening	
oPEo	Opening under way	
oPEd	Pedestrian opening under way	
CLoS	Closing under way	
StEP	Control unit waiting for commands after a start impulse with step-by-step operation	
StoP	Intervened STOP or RESET input during gate operation or detected obstacle with limited reversal duration (P055 SA.07 > 0 or P056 SA.08 > 0)	
Phot	The PHOTO input intervened during the manoeuvre	
-88-	TYPE not defined (see Paragraph 4.2)	
⋈	With the energy-saving function enabled, the display shows the indicated message every 10s	
UoLt	Insufficient voltage. Check the power supply to the control board.	
L L	With the control unit powered but the display switched off completely, the board is in BOOT MODE: Indicates that the firmware is corrupted or is being updated. To restore the firmware, use the DEInstaller app and make sure that the NET-NODE is connected to the correct port. Warning: when the firmware is updated, the board loses all the data (parameters and remote controls) stored in the memory. Make sure that the memory has been backed-up if the data must be restored after the update.	
	All manoeuvres executed during a slowdown stop search procedure (RESP) will appear flashing on the display.	
ERROR MESSAGES		
Mess.	Description	Possible solutions
Er09	Communication with the device connected to the serial ports (COM1, COM2, etc...) absent or interrupted	Check that the connection cable is in good condition and that it is correctly connected to the UART ports between the control unit and the device in use.
Er14	Motor stroke learning failed	Repeat the motor stroke learning procedure (P003 SE.03).
Er15	Motor stroke learning request	Perform learning of the motor stroke (P003 SE.03) before carrying out any other operation.
Er20	An attempt to programme the board is being carried out when a NET-LINK device is connected	Disconnect the power supply, disconnect the NET-LINK device from the communication port and restore the power supply.
Er21	Programming lock	Unlock the programming by modifying parameter P008 SY.07=000 Reset the control unit to remove the programming lock. Warning: the reset also implies the return to the default values for the entire parameter list.
Er30	Message appearing on the display at the start of each manoeuvre that indicates the activation or a fault of the STOP or RESET input	Verify whether the command operates correctly.
Er31	Message appearing on the display at the start of each manoeuvre that indicates the activation or a fault of the PHOTO_1 input	Verify that the safety devices installed work properly.
Er32	Message appearing on the display at the start of each manoeuvre that indicates the activation or a fault of the PHOTO_2 input	
Er33	Message appearing on the display at the start of each manoeuvre that indicates the activation or a fault of the SAFETY_1 input	
Er34	Message appearing on the display at the start of each manoeuvre that indicates the activation or a fault of the SAFETY_2 input	
Er71	Possible fault on encoder channel 1	Verify that the encoder is properly connected. If the wiring is correct, perhaps encoder channel 1 is faulty. Set the control unit to function with only 1 encoder channel (P012 SY.04=001), but it is necessary to invert the CH1-CH2 wires of the encoder. If the error persists, replace the electric motor.

E-72	Possible fault on encoder channel 2	Verify that the encoder is properly connected. If the wiring is correct, perhaps encoder channel 2 is faulty. Set the control unit to function with only 1 encoder channel (P012 SY.04=001). Warning: The encoder's precision will be reduced
E-93 5555	Possible failure of the internal components of the control unit.	Disconnect and restore power to the control unit. If the error persists, replace the control unit.
E-94	The motor has exceeded 10s of continuous operation with zero speed.	/
E-95	The motor has exceeded 5 minutes of continuous operation.	/

7 COMPLETE PARAMETER LIST

The list below helps determine if the parameter you want to interact with is available for the set TYPE of operation.

For the complete list of functions and selectable values, refer to the detailed parameter list (page 42).

Menù		Description	Used in	
"NET"	"EVO"		TYPE 00	TYPE 04
P003 ▶	SE.03	Motor stroke learning	•	•
P004 ▶	RA.01	Deletion of all remote controls in memory	•	•
P005 ▶	RA.02	Learning of remote controls	•	•
P006 ▶	RA.03	Searching and deleting a remote control	•	•
P007 ▶	SY.05	Restoring the operating parameters (excluding IN / OUT)	•	•
P008 ▶	SY.07	Programming access lock	•	•
P009 ▶	RA.09	Memorised remote controls	•	•
P010 ▶	IO.00	Restoration of the "I/O" (input/output) settings	•	•
P011 ▶	SY.00	TYPE set	•	•
P012 ▶	SY.04	Encoder type	•	•
P013 ▶	TE.01	Control board input status visualization	•	•
P014 ▶	TE.02	Expansion board input status visualization	•	•
P017 ▶	IO.01	INPUT 1	•	•
P018 ▶	IO.02	INPUT 2	•	•
P019 ▶	IO.03	INPUT 3	•	•
P020 ▶	IO.04	INPUT 4	•	•
P023 ▶	RA.05	Remote control button 1	•	•
P024 ▶	RA.06	Remote control button 2	•	•
P025 ▶	RA.07	Remote control button 3	•	•
P026 ▶	RA.08	Remote control button 4	•	•
P027 ▶	RA.04	Radio coding	•	•
P028 ▶	SY.01	Motor type	•	•
P029 ▶	SY.03	Encoder used	•	•
P031 ▶	MO.01	Slow down speed in opening	•	•
P032 ▶	MO.02	Maximum speed in opening	•	•
P033 ▶	MO.07	Maximum speed in closing	•	•
P034 ▶	MO.06	Slow down speed in closing	•	•
P035 ▶	MO.05	Slow down duration in opening	•	•
P036 ▶	MO.10	Slow down duration in closing	•	•

P037	►	MO.13	Motor 1 obstacle detection sensitivity in opening	•	•
P038	►	MO.14	Motor 1 obstacle detection sensitivity in closing	•	•
P041	►	TI.01	Automatic closing time (TCA)	•	•
P042	►	TI.02	Pedestrian automatic closing time (TCA_PED)	•	•
P043	►	MO.12	Percentage of pedestrian opening	•	•
P044	►	TI.03	Opening pre-manoeuve flashing time	•	•
P047	►	EX.01	Condominium function	•	•
P049	►	EX.06	“STEP-BY-STEP” enabling	•	•
P050	►	SA.01	PHOTO_1	•	•
P051	►	SA.02	PHOTO_2	•	•
P052	►	IO.32	OUTPUT 2	•	•
P053	►	MO.17	RESP and stop management	•	•
P054	►	/	NOT USED		
P055	►	SA.07	Reversal due to obstacle during opening	•	•
P056	►	SA.08	Inversione su ostacolo in chiusura	•	•
P057	►	EX.07	Manual unlocking facilitation with gate closed	•	•
P058	►	MO.21	Stop margin at opening	•	•
P059	►	MO.22	Stop margin at closing	•	•
P061	►	EX.10	Energy Saving	•	•
P062	►	EX.04	Electric lock activation	•	•
P063	►	SE.04	Motor running direction	•	•
P064	►	TI.08	ELOCK_M1 Time	•	•
P065	►	SY.06	Maintenance manoeuvre counter	•	•
P066	►	IO.31	OUTPUT 1	•	•
P067	►	SA.03	SAFETY_1	•	•
P068	►	SA.04	SAFETY_2	•	•
P070	►	MO.19	Initial surge duration on opening	•	•
P071	►	SA.10	Safety device auto-test	•	•
P072	►	EX.11	SAS function	•	•
P073	►	EX.17	Forced hold-to-run	•	•
P074	►	SA.11	PHOTO_1 inhibition		•
P076	►	SY.09	Save position		•
P077	►	MO.28	Anti-tampering brake		•
P084	►	IO.20	Operation of the button [↑]		•
P085	►	IO.21	Operation of the button [↓]		•
P093	►	IO.60	STATUS 1 function	•	•
P094	►	IO.61	STATUS 2 function	•	•
P095	►	SY.08	NET-EXP enabling	•	•
P096	►	IO.41	EXP_INPUT 1	•	•
P097	►	IO.42	EXP_INPUT 2	•	•
P098	►	IO.43	EXP_INPUT 3	•	•
P099	►	IO.44	EXP_INPUT 4	•	•
P100	►	IO.45	EXP_INPUT 5	•	•
P101	►	IO.46	EXP_INPUT 6	•	•
P102	►	IO.51	EXP_OUTPUT 1	•	•

P103	▶	IO.52	EXP_OUTPUT 2	•	•
P104	▶	IO.53	EXP_OUTPUT 3	•	•
P105	▶	IO.54	EXP_OUTPUT 4	•	•
P106	▶	IO.55	EXP_OUTPUT 5	•	•
P107	▶	IO.56	EXP_OUTPUT 6	•	•
P108	▶	IO.57	EXP_OUTPUT 7	•	•
P109	▶	IO.58	EXP_OUTPUT 8	•	•
P110	▶	RA.10	Learning via hidden button	•	•
P111	▶	SY.12	Type communication port U1/S1	•	•
P113	▶	SY.14	Type communication port U3/S3	•	•
P114	▶	MO.03	Acceleration distance on opening	•	•
P115	▶	MO.04	Opening speed transition space	•	•
P116	▶	MO.08	Acceleration distance on closing	•	•
P117	▶	MO.09	Closing speed transition space	•	•
P118	▶	MO.11	Stop Intensity	•	•
P119	▶	MO.18	Double-Phase Acceleration		•
P120	▶	MO.20	Initial surge duration on closing	•	•
P125	▶	SA.05	“CLOSE IMMEDIATELY” activation	•	•
P126	▶	SA.06	“CLOSE IMMEDIATELY” behaviour	•	•
P127	▶	SA.09	Automatic closing time (TCA) with EMERGENCY_IN	•	•
P128	▶	SA.12	Automatic closing (TCA) after obstacle during closing	•	•
P129	▶	TI.04	Closing pre-manoeuve flashing time	•	•
P130	▶	TI.07	COURTESY Time	•	•
P138	▶	TI.16	AUX_1 Time	•	•
P139	▶	TI.17	AUX_2 Time	•	•
P141	▶	TI.19	Failed closing alarm time	•	•
P142	▶	TI.20	Delay on “CLOSE IMMEDIATELY”	•	•
P145	▶	EX.08	Manual unlocking facilitation with gate open	•	•
P146	▶	EX.12	Automatic close time (TCA) reload	•	•
P147	▶	EX.13	Suspend the automatic close time (TCA)	•	•
P151	▶	EX.18	Displays the manoeuvres once the board is powered	•	•
P153	▶	EX.20	MANEUVER_INHIBITION function	•	•
P154	▶	TE.03	Manoeuvre counter status visualization	•	•
P155	▶	TE.05	Automatic opening time (TAA)	•	•
P156	▶	TE.11	Alternative view of the parameter lists	•	•

8 DETAILED PARAMETER LIST

The following is a detailed list of parameters divided into subcategories.

 SETUP: Parameters that manage learning of the motors and the positioning of the doors.	 RADIO: Parameters that manage the remote controls and the functions associated with them.	 SYSTEM: Parameters that manage essential functions for the system's operation.
 IN/OUT: Parameters that manage the inputs/ outputs of connected devices.	 MOVEMENT: Parameters that manage the movement, speed and force values of the doors.	 SAFETIES: Parameters that manage the safety devices and the functions associated with them.
 TIMES: Parameters that manage functions associated with the operating times.	 EXTRA: Parameters that include specific functions for certain types of automations or systems.	 TEST: Parameters that include diagnostics functions.

SETUP parameters ()

			Default	
			TYPE 00	TYPE 04
P003	SE.03	Motor stroke learning		
		Start the stroke learning procedure: Press [OK] once: CL-1 will be displayed; CL-1: bring the motor to the desired closing point using the [+] and [-] buttons. Confirm the position by pressing [OK] until OP-1 appear on the display; OP-1: bring the motor to the desired opening point using the [+] and [-] buttons. Confirm the position by pressing [OK] until CLOS flashes then release the button. The CLOS message may remain on the display for a variable time, after which the motor will start an automatic maneuver (close-open) to memorize the stroke and forces. At the end of the procedure P003 SE.03 reappears. Attention: If there are wired limit switches, the motor will stop automatically during the positioning phase when a limit switch is reached. It will however be necessary to confirm the position with the [OK] button to continue.		
P063	SE.04	Motor running direction	000	000
		Motion direction reversal: If=1 automatically reverses the motor's open/close outputs, so the wiring does not have to be modified manually if the gearmotor is installed in the inverted position with respect to the standard. Warning: By changing this parameter the limit switches will NOT be inverted if present. Warning: When this parameter is modified, a new motor stroke learning procedure must be carried out (P003 SE.03).	• 000: Standard installation • 001: Reversed installation	

RADIO parameters ()

			Default	
			TYPE 00	TYPE 04
P004	rA.01	Deletion of all remote controls in memory		
		- Briefly press [OK] once: CANC will appear flashing; - Press and hold [OK]: the word CANC will become steady; After roughly 5 s, when P004 rA.01 reappears, release the [OK] button (all remote controls will have been deleted).		
P005	rA.02	Learning of remote controls		
		- Press [OK] once: LEAR will appear for roughly 10 s; - while the word LEAR is displayed, press any button of the remote control to be memorised; - If the memorisation succeeds, the display will show the position assigned to the new saved remote control (e.g. r000, r001, etc.).		
P006	rA.03	Searching and deleting a remote control		
		- briefly press [OK] once; - use the [+] and [-] buttons to reach the position assigned to the remote control that must be deleted; - press [OK] and hold it for roughly 5s until "r- -" appears on the display (the selected remote control has been deleted); - wait for the procedure to end (P006 rA.03 will appear once again). Warning: if there are no remote controls memorised, by accessing the parameter the word "no-r" will appear on the display.		
P027	rA.04	Radio coding	000	000
		• 000: HCS FIXED CODE • 001: HCS ROLLING CODE • 002: DIP SWITCH (HT12) • 003: DART Warning: if the type of encoding must be varied and only if the memory already contains remote controls with different codes, the memory deletion procedure (P004 rA.01) must be carried out AFTER setting the new code.		

P023	rA.05	Remote control button 1	001	001
P024	rA.06	Remote control button 2	000	000
P025	rA.07	Remote control button 3	000	000
P026	rA.08	Remote control button 4	000	000
		• 000: NONE • 001: START • 002: PED • 003: OPEN • 004: CLOSE	• 005: Not Used • 006: Not Used • 007: OPEN_INT • 008: OPEN_EXT • 009: OPEN_STOP	• 010: AUX_1_IN • 011: AUX_2_IN • 012: COURTESY_IN
P009	rA.09	Memorised remote controls		
		By selecting this parameter the display will show the number of remote controls saved in the memory. Use the [OK] button to visualise them.		
P110	rA.10	Learning via hidden button	001	001
		Can be used to enable programming of new remote controls via radio using the hidden button. • 000: programming deactivated; • 001: programming enabled, which nonetheless remains disabled if the control unit is locked using the installer code or the HCS remote control (see P008 SY.07); • 002: always enabled, even with the control unit locked; Note: The hidden-button function is nonetheless always disabled with NET-NODE, NET-BOX or NET-COM connected (ER20 on the display).		

SYSTEM parameters (SYSE)

			Default	
			TYPE 00	TYPE 04
P011	SY.00	TYPE set		
		Allows for viewing the TYPE and the firmware currently being used. - Press the [OK] button: the display briefly shows the TYPE (e.g.: "-00-") followed by the firmware version (e.g.: "00.01").		
P028	SY.01	Motor type	010	000
		TYPE 00 • 010: LIVI 4/24N - LIVI 4/24X TYPE 04 • 000: ADVANCE - ADVANCE/N - SPACE - SPACE/N • 001: SPACE/L • 002: ADVANCE/XL - SPACE/XL • 003: SPACE/XXL - SPACE/N/XXL • 004: SPACE/N/L BOOST • 005: SPACE/N/XL - ADVANCE/N/XL • 006: SPAZIO 702S • 007: SPAZIO 703S		
P030	SY.02	Not Used		
P029	SY.03	Encoder used	000	000
		• 000: motors with wired encoder • 001: motors without wired encoder (virtual encoder)		
P012	SY.04	Encoder type	001	002
		• 001: Encoder 1 channel • 002: Encoder 2 channels		
P007	SY.05	Restoring the operating parameters (excluding IN / OUT)		
		Restores the parameters of the control unit to the default values for the set TYPE. Does not modify the values of the inputs and outputs. Enter in P007 SY.05 with the [OK] button: [dEF1] will appear flashing. Press and hold the [OK] button until [dEF1] stops flashing; Release the [OK] button.		
P065	SY.06	Maintenance manoeuvre counter	000	000
		If=0 it resets the counter and disables the intervention request, if>0 it indicates the number of manoeuvres (x500) to be completed before the control unit performs a pre-flash of 4 additional seconds to signal the need to carry out a maintenance intervention. E.g.: If P065 SY.06=050, the number of manoeuvres = 50x500=25000 Warning: Before setting a new value for the maintenance manoeuvre counter, the latter must be reset by setting P065 SY.06=0 and only later P065 SY.06= "new value".		
			• 000: Maintenance request disabled • >000: Number of manoeuvres (x500) for maintenance request (1.....255)	

P008	SY.07	Programming access lock		
		Access to the control unit programming can be locked, so that the parameters cannot be modified through the display and the buttons. After entering P008 SY.07 two possible messages will appear: • FREE: no lock active • BLOC-PROG: lock active The lock can be set in two modes: • Insertion with remote control through HCS encoding: within P008 SY.07, press the remote control button to lock/unlock. • Installer code setting through NET-LINK devices. Warning: In the absence of the lock code, the control unit can still be unlocked by changing the TYPE or by restoring the operating parameters (P007 SY.05).		
P095	SY.08	NET-EXP enabling	000	000
		NET-EXP expansion board enabling: Warning: by default the expansion board is disabled. Warning: if a settings default is performed, remember to set the parameters correctly.	• 000: Disabled • 001: Enabled	
P076	SY.09	Save position	/	001
		Saving of the gate's position in the memory in case of a power outage. Warning: To complement this parameter, also check the "RESP and stop management" function (P053 Mo.17).	• 000: RESP_ON = At the switch-off, the gate's position is not saved in the memory. The RESP is performed at the subsequent manoeuvre. • 001: RESP_OFF = At the switch-off, the gate's position is saved in the memory. The RESP is not carried out.	
P016	SY.10	Not Used		
P015	SY.11	Not Used		
P111	SY.12	Type communication port U1/S1	000	000
		Activates the serial communication in U1 port (if available on the board) or in S1 (available in art. 485/3-EXP). Warning: It is not possible to occupy both ports at the same time.	• 000: U1_UART • 001: S1_RS485	
P112	SY.13	Not Used		
P113	SY.14	Type communication port U3/S3	000	000
		Activates the serial communication in U3 port (if available on the board) or in S3 (available in art. 485/3-EXP - NET-NODE 2.0). Warning: It is not possible to occupy both ports at the same time.	• 000: U3_UART • 001: S3_RS485	

INPUT / OUTPUT parameters (I/O)

			Default	
			TYPE 00	TYPE 04
P010	Io.00	Restoration of the "I/O" (input/output) settings		
		Restores the inputs and outputs to the default values for the set TYPE (on the control unit and NET-EXP). Enter in P010 Io.00 with the [OK] button: dEF2 will appear flashing; press and hold the [OK] button until dEF2 stops flashing; Release the [OK] button.		
P017	Io.01	INPUT 1	021	021
P018	Io.02	INPUT 2	001	001
P019	Io.03	INPUT 3	017	000
P020	Io.04	INPUT 4	019	000
		• 000: NONE • 001: START • 002: PED • 003: OPEN • 004: CLOSE • 005: OPEN_PM • 006: CLOSE_PM • 007: OPEN_INT • 008: OPEN_EXT • 009: OPEN_STOP • 010: AUX_1_IN • 011: AUX_2_IN • 012: COURTESY_IN • 013: Not Used • 014: Not Used • 015: Not Used • 016: STOP / SAS_INPUT • 017: SWO_1 • 018: Not Used • 019: SWC_1 • 020: Not Used • 021: PHOTO_1 • 022: PHOTO_2 • 023: SAFETY_1 • 024: SAFETY_1_8k2 • 025: SAFETY_2 • 026: SAFETY_2_8k2 • 027: SAFETY_INHIBITION • 028: EMERGENCY_IN • 029: Not Used • 030: RESET • 031: MANEUVER_INHIBITION		

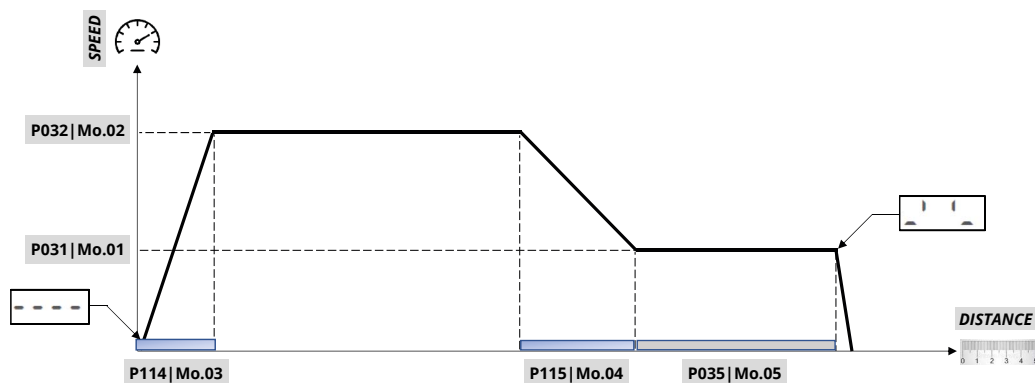
P084	Io.20	Operation of the button [↑]			/	003
P085	Io.21	Operation of the button [↓]			/	004
		• 000: NONE • 001: START • 002: PED • 003: OPEN • 004: CLOSE	• 005: OPEN_PM • 006: CLOSE_PM • 007: OPEN_INT • 008: OPEN_EXT • 009: OPEN_STOP	• 010: AUX_1_IN • 011: AUX_2_IN • 012: COURTESY_IN		
P066	Io.31	OUTPUT 1			011	011
P052	Io.32	OUTPUT 2			009	009
		• 000: NONE • 001: 24V • 002: 24V_TEST • 003: ELOCK_M1 • 004: ELOCK_INV_M1 • 005: ELOCK_BOOST_M1 • 006: Not Used • 007: Not Used • 008: Not Used • 009: WARN_FIX / SAS_OUTPUT • 010: WARN_INT • 011: FLASH_FIX • 012: FLASH_INT	• 013: COURTESY • 014: EBRAKE_M1_N.ON • 015: Not Used • 016: EBRAKE_M1_N.OFF • 017: Not Used • 018: MINUTERIE • 019: ALARM • 020: AUX_1_OUT_INPULS • 021: AUX_1_OUT_STEP • 022: AUX_1_OUT_TEMP • 023: AUX_2_OUT_INPULS • 024: AUX_2_OUT_STEP • 025: AUX_2_OUT_TEMP	• 026: TRAFFIC_LIGHT_INT • 027: TRAFFIC_LIGHT_EXT • 028: STATUS_1 • 029: STATUS_2 • 030: Not Used • 031: Not Used • 032: Not Used • 033: Not Used • 034: Not Used		
P096	Io.41	EXP_INPUT 1			000	000
P097	Io.42	EXP_INPUT 2			000	000
P098	Io.43	EXP_INPUT 3			000	000
P099	Io.44	EXP_INPUT 4			000	000
P100	Io.45	EXP_INPUT 5			000	000
P101	Io.46	EXP_INPUT 6			000	000
		• 000: NONE • 001: START • 002: PED • 003: OPEN • 004: CLOSE • 005: OPEN_PM • 006: CLOSE_PM • 007: OPEN_INT	• 008: OPEN_EXT • 009: OPEN_STOP • 010: AUX_1_IN • 011: AUX_2_IN • 012: COURTESY_IN • 013: Not Used • 014: Not Used • 015: Not Used	• 016: STOP / SAS_INPUT • 017: SWO_1 • 018: Not Used • 019: SWC_1 • 020: Not Used • 021: PHOTO_1 • 022: PHOTO_2 • 023: SAFETY_1	• 024: SAFETY_1_8k2 • 025: SAFETY_2 • 026: SAFETY_2_8k2 • 027: SAFETY_INHIBITION • 028: EMERGENCY_IN • 029: Not Used • 030: RESET • 031: MANEUVER_INHIBITION	
P102	Io.51	EXP_OUTPUT 1			000	000
P103	Io.52	EXP_OUTPUT 2			000	000
P104	Io.53	EXP_OUTPUT 3			000	000
P105	Io.54	EXP_OUTPUT 4			000	000
P106	Io.55	EXP_OUTPUT 5			000	000
P107	Io.56	EXP_OUTPUT 6			000	000
P108	Io.57	EXP_OUTPUT 7			000	000
P109	Io.58	EXP_OUTPUT 8			000	000
		• 000: NONE • 001: 24V • 002: 24V_TEST • 003: ELOCK_M1 • 004: ELOCK_INV_M1 • 005: Not Used • 006: Not Used • 007: Not Used • 008: Not Used • 009: WARN_FIX / SAS_OUTPUT • 010: WARN_INT • 011: FLASH_FIX • 012: FLASH_INT	• 013: COURTESY • 014: EBRAKE_M1_N.ON • 015: Not Used • 016: EBRAKE_M1_N.OFF • 017: Not Used • 018: MINUTERIE • 019: ALARM • 020: AUX_1_OUT_INPULS • 021: AUX_1_OUT_STEP • 022: AUX_1_OUT_TEMP • 023: AUX_2_OUT_INPULS • 024: AUX_2_OUT_STEP • 025: AUX_2_OUT_TEMP	• 026: TRAFFIC_LIGHT_INT • 027: TRAFFIC_LIGHT_EXT • 028: STATUS_1 • 029: STATUS_2 • 030: Not Used • 031: Not Used • 032: Not Used • 033: Not Used • 034: Not Used		
P093	Io.60	STATUS 1 function			000	000
P094	Io.61	STATUS 2 function			000	000
		• 000: NONE • 001: MAINTENANCE • 002: PHOTO • 003: STOP • 004: OBSTACLE • 005: EMERGENCY	• 006: CLOSED_M1 • 007: OPENED_M1 • 008: Not Used • 009: Not Used • 010: RESP_FIX • 011: RESP_INT	• 012: Not Used • 013: Not Used • 014: Not Used • 015: Not Used • 016: CLOSURE_FAILED • 017: SECURITY		

MOVEMENT parameters (ΠΟΛΕ)			Default	
			TYPE 00	TYPE 04
P031	Mo.01	Slow down speed in opening	040	055
		Adjustment of the motor speed during the slow phase on opening.	5%.....100%	
P032	Mo.02	Maximum speed in opening	100	100
		Adjustment of the motor speed during the opening movement.	5%.....100%	
P114	Mo.03	Acceleration distance on opening	010	010
		Refers to a distance defined in % of the total path within which the starting acceleration is created from an opening command. A small value corresponds to a very high ramp.	0%.....30%	
P115	Mo.04	Opening speed transition space	010	010
		It expresses a space defined in % of the total stroke within which the automation must slow down to the deceleration speed. The end of this section is calculated starting from the point where the P035 Mo.05 space starts.	0%.....30%	
P035	Mo.05	Slow down duration in opening	010	010
		Refers to a space defined in % on the total path. It represents the last section completed by the automation during the opening manoeuvre.	0%.....30%	
P034	Mo.06	Slow down speed in closing	040	020
		Adjustment of the motor speed during the slowdown phase on closing.	5%.....100%	
P033	Mo.07	Maximum speed in closing	100	075
		Adjustment of the motor speed during the closing movement.	5%.....100%	
P116	Mo.08	Acceleration distance on closing	010	010
		Refers to a distance defined in % of the total path within which the starting acceleration is created from an closing command. A small value corresponds to a very high ramp.	0%.....30%	
P117	Mo.09	Closing speed transition space	010	010
		It expresses a space defined in % of the total stroke within which the automation must slow down to the deceleration speed. The end of this section is calculated starting from the point where the P036 Mo.10 space starts.	0%.....30%	
P036	Mo.10	Slow down duration in closing	010	010
		Refers to a space defined in % of the total path. It represents the last section completed by the automation during the closing manoeuvre.	0%.....30%	
P118	Mo.11	Stop Intensity	020	020
		Adjusts how quickly the motor drops to 0 (zero) speed in the event of a stop or movement reversal. Affects stops generated by: - movement commands (START, OPEN, CLOSE, PED, etc.); - PHOTO or STOP type input activation. It does not affect stops generated by: - arrival on limit switch or on memorized point; - emergency reversal (internal obstacle detection or SAFETY type input activation). Note: Higher values = faster stop; Low values = slower stop. Warning: To prevent excessive stresses that could compromise the proper functioning of the automation, it is recommended to always take into account the possible inertia of the door, avoiding setting too high values in the presence of important masses.	• 000: Deactivated • >000: Active intensity for the set value (1%.....50%) Attention: With the function deactivated, the motor stops immediately.	
P043	Mo.12	Percentage of pedestrian opening	030	050
		Adjustment of the distance of the pedestrian opening movement.	5%.....100%	
P037	Mo.13	Motor 1 obstacle detection sensitivity in opening	050	040
		Obstacle detection sensitivity adjustment for Motor 1 in opening, based on the amount of force required to trigger the obstacle detection. Higher number is less sensitive. (if = 100% obstacle detection disabled)	1%.....100%	
P038	Mo.14	Motor 1 obstacle detection sensitivity in closing	050	040
		Obstacle detection sensitivity adjustment for Motor 1 in closing, based on the amount of force required to trigger the obstacle detection. Higher number is less sensitive. (if = 100% obstacle detection disabled)	1%.....100%	

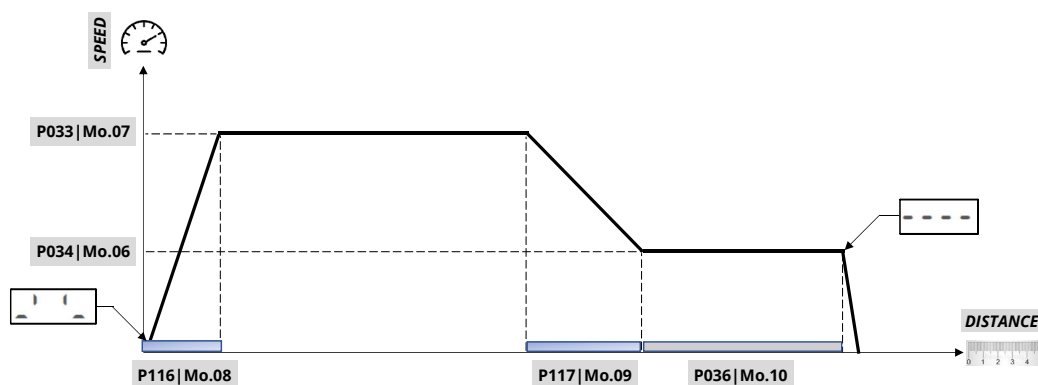
P039	Mo.15	Not Used		
P040	Mo.16	Not Used		
P053	Mo.17	RESP and stop management	002	001
		• 000: In opening and closing, it stops at the memorized points, and in case of RESP, the first operation will be to search for the open stop; • 001: In opening and closing, it stops at the memorized points, and in case of RESP, the first operation will be to search for the closed stop; • 002: Searches for the stop only on opening, while on closing it stops on the memorised point. In case of RESP, the first operation will be in the open position. • 003: Searches for the stop only on closing, while on opening it stops on the memorised point. In case of RESP, the first operation will be in the closed position. • 004: Searches for the stop on closing and on opening. In case of RESP, the first operation will be in the open position. • 005: Searches for the stop on closing and on opening. In case of RESP, the first operation will be in the closed position. Note: The only direction allowed in case of RESP is the set direction, in case of reversals (PHOTO, SAFETY) it stops the movement. Motion in the opposite direction is allowed if an end switch is present in that direction. Note: With wired limit switches, the automation always stops when the limit switch is reached. Note: Enable the search for setbacks only after making the stroke learning (P003 SE.03). This is to prevent the motor from pushing too hard during the mapping of the forces.		
P119	Mo.18	Double-Phase Acceleration	/	000
		If activated, it executes an acceleration ramp until reaching the slow down speed determined by parameter P031 Mo.01 or P034 Mo.06, until the end of the acceleration space (P114 Mo.03 or P116 Mo.08), after which the speed goes to the maximum speed. Note: This parameter can be used to facilitate the impact tests during the closing manoeuvre at the point: H-300mm. (H = height of the sectional door)	• 000: Disabled • 001: Active only in opening • 002: Active only in closing • 003: Active both in opening and closing	
P070	Mo.19	Initial surge duration on opening	000	000
		• 000: Initial surge deactivated (performs a very short, almost imperceptible surge) • 00X: Adjusts the duration of the surge up to 2,5s (X*10 ms)		
P120	Mo.20	Initial surge duration on closing	000	000
		• 000: Initial surge deactivated (performs a very short, almost imperceptible surge) • 00X: Adjusts the duration of the surge up to 2,5s (X*10 ms)		
P058	Mo.21	Stop margin at opening	010	100
		Space at the end of the opening manoeuvre performed with deceleration speed and without reversing onto an obstacle. Expressed in thousandths of the total stroke. (0.....255)		
P059	Mo.22	Stop margin at closing	010	030
		Space at the end of the closing manoeuvre performed with deceleration speed and without reversing over an obstacle. Expressed in thousandths of the total stroke. (0.....255)		
P121	Mo.23	Not Used		
P060	Mo.24	Not Used		
P122	Mo.25	Not Used		
P123	Mo.26	Not Used		
P124	Mo.27	Not Used		
P077	Mo.28	Anti-tampering brake	/	000
		Parameter intended for sectional doors. Allows to counteract any motor movements that are detected when the automation is not in operation. In this case the motor will be activated in the opposite direction to the movement detected, in order to keep the automation position unchanged. Warning: It is NECESSARY to have a double channel encoder wired and enabled (P012 SY.04=002). Warning: If active (P077 Mo.28 = 001,002,003) it is NECESSARY to set parameter P053 Mo.17 = 002 and install the mechanical stop (art. AB / FM not supplied) on the track in the open position.		

DIAGRAMS FOR THE ADJUSTMENT OF THE MOVEMENT

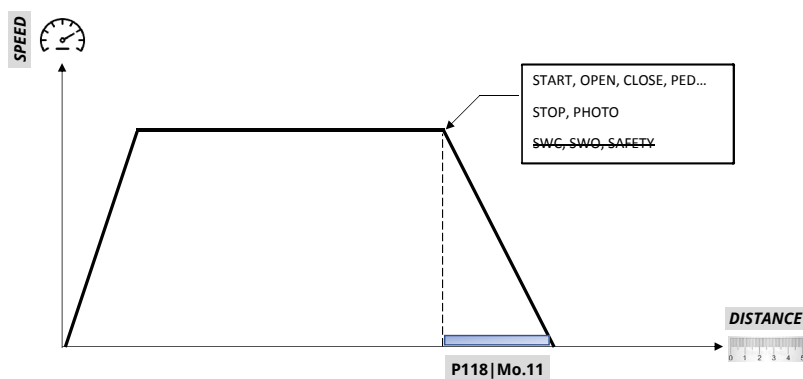
Adjustments in OPENING



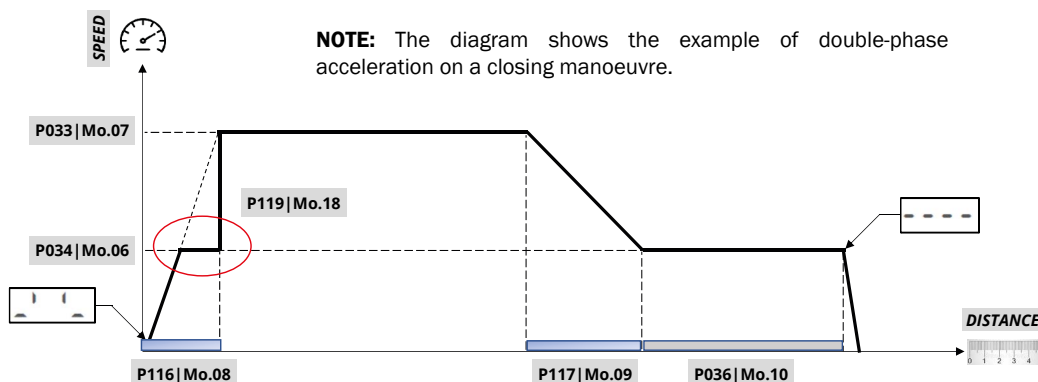
Adjustments in CLOSING



Adjustment of the STOP INTENSITY (P118 | Mo.11)



Double-phase acceleration adjustment (P119 | Mo.18)



SAFETIES parameters (SAFE)

			Default	
			TYPE 00	TYPE 04
P050	SA.01	PHOTO_1	002	002
		• 000: STOP & CLOSE - The photocell input is active before movement; a trigger will prevent movement. During closing a trigger will reverse movement. • 001: ALWAYS - The photocell input is always active. With the gate stationary a trigger will prevent movement. During opening a trigger will stop the gate until trigger removed. During closing a trigger will reverse the movement of the gate. • 002: CLOSE - The photocell input is active only on closing, a trigger will reverse the movement of the gate.		
P051	SA.02	PHOTO_2	000	002
		• 000: STOP & CLOSE - The photocell input is active before movement; a trigger will prevent movement. During closing a trigger will reverse movement. • 001: ALWAYS - The photocell input is always active. With the gate stationary a trigger will prevent movement. During opening a trigger will stop the gate until trigger removed. During closing a trigger will reverse the movement of the gate. • 002: CLOSE - The photocell input is active only on closing, a trigger will reverse the movement of the gate.		
P067	SA.03	SAFETY_1	000	002
		• 000: always enabled • 001: enabled only during closing • 002: enabled only during closing and before starting a closing maneuver • 003: enabled during closing, opening, and before starting a closing maneuver • 004: enabled only during opening • 005: enabled only during opening and before starting an opening maneuver • 006: enabled during opening, closing, and before starting an opening maneuver		
P068	SA.04	SAFETY_2	000	002
		• 000: always enabled • 001: enabled only during closing • 002: enabled only during closing and before starting a closing maneuver • 003: enabled during closing, opening, and before starting a closing maneuver • 004: enabled only during opening • 005: enabled only during opening and before starting an opening maneuver • 006: enabled during opening, closing, and before starting an opening maneuver		
P125	SA.05	"CLOSE IMMEDIATELY" activation	000	000
		• 000: Disabled • 001: Active only for PHOTO_1 • 002: Active only for PHOTO_2 • 003: Active for PHOTO_1 and PHOTO_2		
P126	SA.06	"CLOSE IMMEDIATELY" behaviour	000	000
		Allows for defining the behaviour when a photocell with "close immediately" mode enabled is crossed during an opening manoeuvre. • 000: Completes the full opening manoeuvre first and then closes the gate again with the delay set in P142 tl.20; • 001: Interrupts the manoeuvre and then closes the gate again with the delay set in P142 tl.20.		
P055	SA.07	Reversal due to obstacle during opening	003	003
		Allows for adjusting the duration of the reversal of motion if an obstacle is detected during the opening manoeuvre. • 000: Complete reversal after an obstacle • >000: Duration of the reversal after obstacle (1s.....10s)		
P056	SA.08	Inversione su ostacolo in chiusura	003	003
		Allows for adjusting the duration of the reversal of motion if an obstacle is detected during the closing manoeuvre. • 000: Complete reversal after an obstacle • >000: Duration of the reversal after obstacle (1s.....10s)		
P127	SA.09	Automatic closing time (TCA) with EMERGENCY_IN	000	000
		Describes the behaviour of the TCA in combination with the EMERGENCY_IN function • 000: If TCA is active, the gate re-closes after the TCA; • 001: Even if TCA is active, the gate does not re-close automatically. The first re-closing manoeuvre requires a command. On the successive manoeuvre, the TCA works normally.		
P071	SA.10	Safety device auto-test	000	000
		The function switches off the 24V_TEST input and verifies the opening of the safety device contact before each manoeuvre. Warning: for operation in the "Safety device auto-test" mode, the following elements must be connected: - The transmitters (TX) on the 24V_TEST output; - The receivers (RX) on the 24V output; Moreover, the safety devices must be wired and aligned before the path is learned (P003 SE.03). • 000: Disabled • 001: Enabled		

P074	SA.11	PHOTO_1 inhibition	/	000
		The function inhibits the intervention of the photocells on both the opening and closing manoeuvres, in the zone between the detection point and the complete closing point. During the motor stroke automatic learning manoeuvre (P003 SE.03), the trigger of the photocells during the door closing movement defines the inhibition start point. It is necessary that the photocell to be inhibited is connected as PHOTO_1.	• 000: Inhibition deactivated • 001: Inhibition active (the photocells are always ignored in the zone between the detection point and the closing point)	
P128	SA.12	Automatic closing (TCA) after obstacle during closing	000	000
		In case of full reversal (P056 SA.08=000) during a closing manoeuvre, it determines how the automation will perform the TCA automatic re-closing manoeuvre. Note: valid only for obstacles detected through the SAFETY mode or the anti-crushing function.	• 000: Always performs automatic re-closing • 001: Attempts 1 automatic re-closing • 002: Attempts 2 automatic re-closing manoeuvres • 003: Attempts 3 automatic re-closing manoeuvres • 004: Inhibits automatic re-closing	

TIMES parameters (Ƨ ƧƧ)

			Default	
			TYPE 00	TYPE 04
P041	tl.01	Automatic closing time (TCA)	000	000
		Adjusts the time for automatic closing	• 000: Disabled • >000: Active for the set time (1s.....255s)	
P042	tl.02	Pedestrian automatic closing time (TCA_PED)	000	000
		Adjusts the time for automatic pedestrian closing	• 000: Disabled • >000: Active for the set time (1.....255s) (1.....255min - ONLY TYPE 04)	
P044	tl.03	Opening pre-manoeuve flashing time	000	000
		Adjusts the pre-flashing time before an opening manoeuvre.	(0s.....10s)	
P129	tl.04	Closing pre-manoeuve flashing time	000	000
		Adjusts the pre-flashing time before a closing manoeuvre.	(0s.....10s)	
P045	tl.05	Not Used		
P046	tl.06	Not Used		
P130	tl.07	COURTESY Time	060	060
		Adjusts the time for the COURTESY light.	0s.....255s	
P064	tl.08	ELOCK_M1 Time	002	002
		Adjusts the activation time for the ELOCK_M1 / BOOST_M1 output, or the deactivation time of the ELOCK_INV_M1 output.	(1s.....10s)	
P131	tl.09	Not Used		
P132	tl.10	Not Used		
P133	tl.11	Not Used		
P134	tl.12	Not Used		
P135	tl.13	Not Used		
P136	tl.14	Not Used		
P137	tl.15	Not Used		
P138	tl.16	AUX_1 Time	100	100
		Adjusts the activation time for the AUX_1 output, if timer-controlled.	(0s.....255s)	
P139	tl.17	AUX_2 Time	100	100
		Adjusts the activation time for the AUX_2 output, if timer-controlled.	(0s.....255s)	
P140	tl.18	Not Used		
P141	tl.19	Failed closing alarm time	010	010
		Time used for the STATUS outputs configured as CLOSURE_FAILED: once this time has elapsed, if the automation is not in the closing position, the output is activated. Note: Functions regardless of the presence of the TCA (P041 tl.01) or TCA_PED (P042 tl.02).	(1min.....255min)	

P142	tl.20	Delay on "CLOSE IMMEDIATELY"	001	001
		Allows for setting after how many seconds the automatic re-closing of the "close immediately" function occurs (P125 SA.05).	(1s.....10s)	

EXTRA parameters (EXTRA)

			Default	
			TYPE 00	TYPE 04
P047	EX.01	Condominium function	000	000
		Allows for disabling the opening and closing command inputs during opening and the automatic closing time.	• 000: Disabled • 001: Active only on opening • 002: Active on opening and during automatic close time	
P048	EX.02	Not Used		
P143	EX.03	Not Used		
P062	EX.04	Electric lock activation	000	000
		• 000: Active only at the start of opening • 001: Active only at the start of closing • 002: Active at the start of opening and closing		
P144	EX.05	Not Used		
P049	EX.06	"STEP-BY-STEP" enabling	001	001
		Selection of the "reversal" mode (during the manoeuvre a command impulse reverses the direction of motion) or "step-by-step" (during the manoeuvre a command impulse stops the movement. The subsequent impulse restarts the movement in the opposite direction).	• 000: Reversal • 001: Step-by-Step with START and PED • 002: Step-by-Step with START, PED and OPEN/CLOSE	
P057	EX.07	Manual unlocking facilitation with gate closed	000	000
		If=0, the function is disabled. If≠0, after the detection of the closing stop, 1 motor carries out a brief reversal to release the pressure on it and facilitate manual unlocking. The set value indicates the duration of the reversal.	• 000: Unlocking facilitation deactivated • >000: Unlocking facilitation activated with duration equal to: (1x25ms.....20x25ms) (1x25ms.....40x25ms - ONLY TYPE 00)	
P145	EX.08	Manual unlocking facilitation with gate open	000	000
		If=0, the function is disabled. If≠0, after the detection of the opening stop, 1 motor carries out a brief reversal to release the pressure on it and facilitate manual unlocking. The set value indicates the duration of the reversal.	• 000: Unlocking facilitation deactivated • >000: Unlocking facilitation activated with duration equal to: (1x25ms.....20x25ms) (1x25ms.....40x25ms - ONLY TYPE 00)	
P069	EX.09	Not Used		
P061	EX.10	Energy Saving	000	000
		Function for energy saving when there is an emergency battery kit or a power supply system based on solar panels. When the Energy Saving function is active, the display is off and shows the "-" symbol every 10s. The function can be set to 3 levels: • 000: "Energy saving" not active; • 001: "Energy saving" active: switches off all outputs except for the Vaux 24V fixed output; • 002: "Energy saving" active: switches off all outputs except for the Vaux 24V fixed output; moreover, any STATUS-type outputs are kept active. Warning: with "Energy saving" active the SAS function is not available.		
P072	EX.11	SAS function	000	000
		Activation of the SAS function (only for SAS_OUTPUT outputs): the SAS output is connected to a STOP / SAS_INPUT input of a second control unit, triggering the "bank door" operating mode (opening of the second door is disabled as long as the first door is not fully closed). If this parameter is enabled, following a reset it performs an automatic RESP during which the SAS output does not activate. If limit switches are present and are crushed after a reset, the RESP will not be carried out. Warning: if both leaves are manually unlocked and shifted from the closing position, the interlock condition is created. It will then be necessary to manually close at least one of the two leaves. Warning: the STOP / SAS_INPUT input must be activated by a voltage-free clean contact output, such as that of a NET-EXP expansion card output or a relay. Warning: in case of RESP, the SAS function must necessarily be linked to a search for the reference / stop towards closure. So it works with the values of P053 Mo.17 = 001, 003, 005.	• 000: Disabled • 001: Enabled	

P146	EX.12	Automatic close time (TCA) reload	001	001
		It allows you to define if, with the gate open and TCA active, the TCA is reloaded. The function is valid both for the TCA (P041 tl.01) and for the TCA_PED (P042 tl.02). If = 1, with the gate completely open, an intervention on PHOTO, OPEN, OPEN_INT or OPEN_EXT reloads the TCA. With the gate open in pedestrian mode, an intervention on PHOTO recharges the TCA_PED, while an OPEN, OPEN_INT or OPEN_EXT command causes complete opening.	• 000: Recharge disabled • 001: Recharge enabled	
P147	EX.13	Suspend the automatic close time (TCA)	000	000
		If=0 with the door open and TCA active, the automatic closing always occurs, even if an impulsive STOP command was given. Only a maintained STOP command inhibits the automatic closing of the door; If=1 with the door open and TCA active, an impulse of the STOP command cancels the automatic closing; If=2 with the door open and TCA active, an impulse of the STOP or OPEN command cancels the automatic closing. Note: If active, the parameter also acts on the TCA_PED function (in this case the value 002 acts as 001 as an OPEN command with the gate stopped in pedestrian opening generates a complete opening).	• 000: Disabled • 001: Active with a STOP impulse disables TCA. • 002: Active with a STOP or OPEN impulse disables TCA.	
P148	EX.14	Not Used		
P149	EX.15	Not Used		
P150	EX.16	Not Used		
P073	EX.17	Forced hold-to-run	000	000
		If the function is activated, all inputs configured as OPEN and CLOSE automatically become also OPEN_UP and CLOSE_UP if activated and kept active for at least 5s, if a safety device (photocell and/or sensitive edge) is engaged. This function, therefore, can be used to command the automation even if the safety devices are faulty. If the input is no longer kept activated, the automation will return to the automatic operating mode. This function is not available with SAFETY-type safety devices deactivated with the gate stationary (values 001 and 004 of parameters P067 SA.03 and P068 SA.04). For safety reasons, we suggest NOT using this function if there are clocks connected to inputs configured as OPEN or CLOSE.	• 000: Function disabled. • 001: Function enabled (automatic transition to OPEN_UP/CLOSE_UP with safety devices engaged/faulty if the OPEN/CLOSE commands are maintained)	
P151	EX.18	Displays the manoeuvres once the board is powered	000	000
		Allows for integrating the messages on the display during the control unit's power-on phase, showing the total number of manoeuvres completed. Warning: The function's activation, and the resulting visualization of new information on the display, causes the control unit to start more slowly	• 000: Visualization of total manoeuvres disabled • 001: Visualization of total manoeuvres enabled	
P152	EX.19	Not Used		
P153	EX.20	MANEUVER_INHIBITION function	000	000
		If an input is set as MANEUVER_INHIBITION it behaves like an NC contact which, if open, inhibits the maneuver.	• 000: if open, it inhibits opening commands but allows closing commands. • 001: if open, it inhibits opening and closing commands.	

TEST parameters (TEST)

TEST parameters (TEST)			Default																							
			TYPE 00	TYPE 04																						
P013	tE.01	Control board input status visualization																								
		Allows for viewing the status of inputs for the control board. An off segment corresponds to an open segment, while an on segment is associated with a closed contact.																								
		1 5 9 2 6 10 3 7 11 4 8 12 <table><tr><td>1</td><td>IN_1</td><td>4</td><td>IN_4</td><td>7</td><td>IN_7</td><td>10</td><td>IN_10</td></tr><tr><td>2</td><td>IN_2</td><td>5</td><td>IN_5</td><td>8</td><td>IN_8</td><td>11</td><td>IN_11</td></tr><tr><td>3</td><td>IN_3</td><td>6</td><td>IN_6</td><td>9</td><td>IN_9</td><td>12</td><td>IN_12</td></tr></table>	1	IN_1	4	IN_4	7	IN_7	10	IN_10	2	IN_2	5	IN_5	8	IN_8	11	IN_11	3	IN_3	6	IN_6	9	IN_9	12	IN_12
1	IN_1	4	IN_4	7	IN_7	10	IN_10																			
2	IN_2	5	IN_5	8	IN_8	11	IN_11																			
3	IN_3	6	IN_6	9	IN_9	12	IN_12																			
P014	tE.02	Expansion board input status visualization																								
		Allows for viewing the status of inputs for the expansion board. An off segment corresponds to an open segment, while an on segment is associated with a closed contact.																								
		1 2 3 4 5 6 <table><tr><td>1</td><td>EXP_IN_1</td><td>4</td><td>EXP_IN_4</td></tr><tr><td>2</td><td>EXP_IN_2</td><td>5</td><td>EXP_IN_5</td></tr><tr><td>3</td><td>EXP_IN_3</td><td>6</td><td>EXP_IN_6</td></tr></table>	1	EXP_IN_1	4	EXP_IN_4	2	EXP_IN_2	5	EXP_IN_5	3	EXP_IN_3	6	EXP_IN_6												
1	EXP_IN_1	4	EXP_IN_4																							
2	EXP_IN_2	5	EXP_IN_5																							
3	EXP_IN_3	6	EXP_IN_6																							
P154	tE.03	Manoeuvre counter status visualization																								
		Allows for viewing the total number of manoeuvres completed by the control unit. The display will show the word “tCYC” with the relative value, followed by the word “MULT” with the relative value of the multiplier. To calculate the number of manoeuvres, the two values must be multiplied. E.g.: tCYC=120, MULT=10; 120x10=1200 manoeuvres completed																								
P075	tE.04	Not Used																								
P155	tE.05	Automatic opening time (TAA)	000	000																						
		Adjusts the time for automatic opening TAA	• 000: Disabled • >000: Active for the set time (1s.....255s)																							
P156	tE.11	Alternative view of the parameter lists	000	000																						
		It allows to alternate the visualization of the lists of parameters between NET series and EVO series.	• 000: NET series • 001: EVO series																							

9 EVO MENU DISPLAY

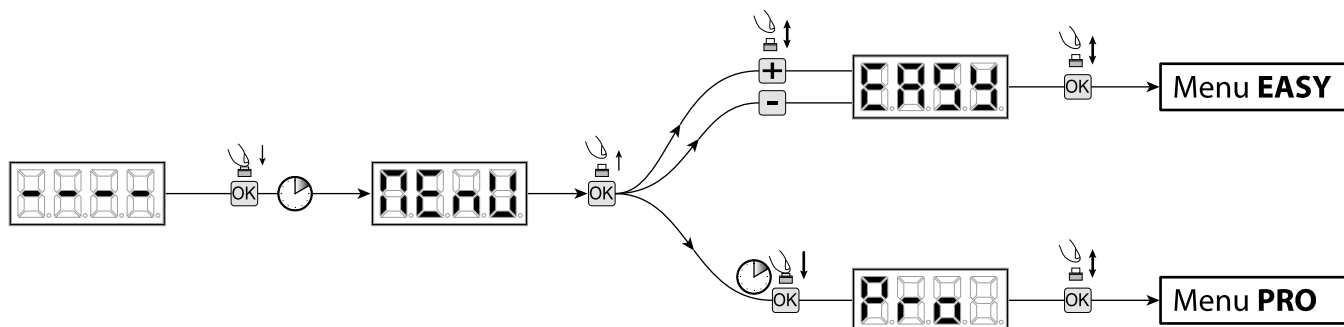
If you want to utilize parameter visualization by subcategories, you can change the “NET” style (set by default) to the “EVO” style.

NOTE: Changing the display style is reversible and not binding.

9.1 Access and navigation with EVO menu

NOTE: Make sure parameter P156|TE.11 is set to 001.

1. With the door stopped, press and hold the [OK] button;
2. When the word **"MENU"** appears; Release the [OK] key;
3. Follow the specific procedures for entering the **"EASY"** or **"PRO"** menus.

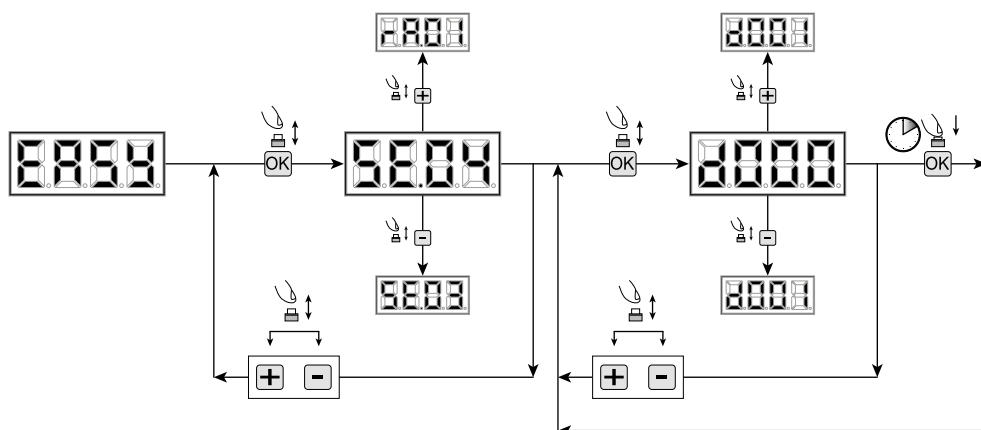


9.2 Navigation in the “EASY” menu

The “**EASY**” menu includes a selection of basic parameters used to start the automation. The list of parameters displayed in the EASY menu varies in relation to the set TYPE.

NOTE: The list of parameters of the “EASY” menu is integrated by any modified parameter (with respect to its DEFAULT value) in the “PRO” menu. This allows for having a list of all the parameters being used.

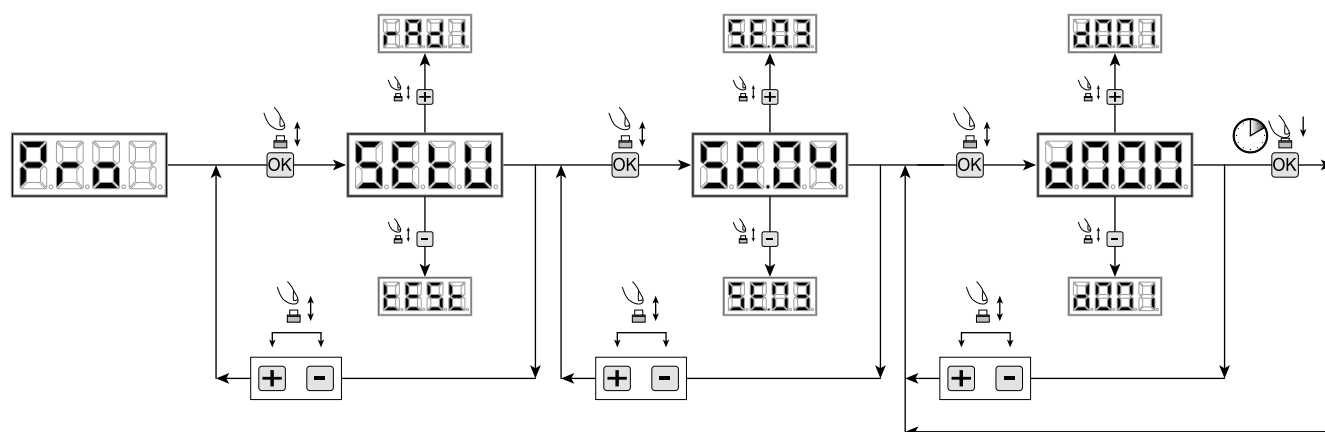
1. Make sure that the word **"EASY"** is visible on the display; Confirm by pressing the [OK] button;
2. Scroll the list of parameters with the [+] or [-] button and confirm the selection by pressing the [OK] button;
3. Modify the parameter with the desired value and confirm with the [OK] button;
4. Simultaneously press the [+] and [-] buttons to exit the menu.



9.3 Navigation in the “PRO” menu

The “**PRO**” menu includes the full list of all available parameters grouped according to sub-categories.

1. Make sure that the word “**PRO**” is visible on the display; Confirm by pressing the [OK] button;
2. Scroll the list of parameter categories with the [+] or [-] button and confirm the selection by pressing the [OK] button;
3. Scroll the list of parameters with the [+] or [-] button and confirm the selection by pressing the [OK] button;
4. Modify the parameter with the desired value and confirm with the [OK] button;
5. Simultaneously press the [+] and [-] buttons to exit the menu.



10 “EASY” PARAMETERS LIST

NOTE When a parameter of the PRO list is changed from its default value, it will appear in the EASY list.

TYPE 00 - (SLIDING GATES)

				Default
P003	SE.03	Motor stroke learning		-
P063	SE.04	Motor running direction		000
P005	RA.02	Learning of remote controls		-
P027	RA.04	Radio coding		000
P023	RA.05	Remote control button 1		001
P024	RA.06	Remote control button 2		000
P019	IO.03	INPUT 3		017
P020	IO.04	INPUT 4		019
P041	TI.01	Automatic closing time (TCA)		000
P042	TI.02	Pedestrian automatic closing time (TCA_PED)		000
P013	TE.01	Control board input status visualization		-
P156	TE.11	Alternative view of the parameter lists		000

TYPE 04 - (SECTIONAL DOORS)

				Default
P003	SE.03	Motor stroke learning		-
P005	RA.02	Learning of remote controls		-
P027	RA.04	Radio coding		000
P023	RA.05	Remote control button 1		001
P024	RA.06	Remote control button 2		000
P041	TI.01	Automatic closing time (TCA)		000
P013	TE.01	Control board input status visualization		-
P156	TE.11	Alternative view of the parameter lists		000

11 COMMISSIONING

The testing operation is essential in order to verify the correct installation of the system. **DEA System** wants to summarize the proper testing of all the automation in 4 easy steps:

- Make sure that you comply strictly as described in paragraph "WARNINGS SUMMARY";
- Test the opening and closing making sure that the movement of the leaf match as expected. We suggest in this regard to perform various tests to assess the smoothness of the gate and defects in assembly or adjustment;
- Ensure that all safety devices connected work properly;
- Perform the measurement of impact forces in accordance with the standard EN12445 to find the setting that ensures compliance with the limits set by the standard EN12453.

12 PRODUCT DISPOSAL

DISASSEMBLY

The automation unit must be dismantled by qualified personnel, in accordance with the current accident prevention and safety regulations, and with reference to the installation instructions, but in the reverse order. Before initiating the disassembly operations, disconnect the electrical power and make sure it cannot be reconnected.

DISPOSAL

The automation unit must be disposed of in accordance with the current local and national waste disposal regulations. The product (or its individual parts) must not be disposed of together with other household waste materials.



WARNING In compliance with EU Directive 2012/19/EU on waste electrical and electronic equipment (WEEE), this electrical product should not be treated as municipal mixed waste. Please dispose of the product and bring it to the collection for an appropriate local municipal recycling.